Logic

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Chapter 1

Atomes

1.1 Basic tools

we start the book by building logic step by step.block by block in first it might seem a little annoying but I advice you to be patient.book will show its power soon. like euclidian geometry this book has undefined concepts. I tried to explain them for you to understand.but they don't have any definition.

Undefined 1.1.1. ***Urelements*** *are the most fundamental entities in a logical. They cannot be constructed, defined, or described in terms of other things. Their existence is primitive—accepted without internal structure or relational properties. Urelements simply are.*

In this book, we will denote individual urelements using lowercase letters u, v, w,...

Undefined 1.1.2. ***Type*** *parallel lines used to show that an object exists (you will undrestnd what is object soon.) (we will show you how to use them).*

1. u

2. v

3. w

Rule 1.1.1. ***fusion*** *it's the Rule that explain how sets made. (this rule is schema.it means it works everywhere for every object.)*

$m.$	u_0	
	$\vdots$	
$m+n.$	u_n	
$o.$		$\{u_0, u_1 ... u_n\}$

Undefined 1.1.3. ***Set*** *we can't say what exactly sets are but we know sets can be made by fusion (if "a" is a set we write "set(a)" from now we denote sets with uppercase letters like A, B, C, ...).*

definition 1.1.1. ***Object*** *urelements and sets.(we write obj instead of object). (from now we denote objects with lowercase letters like a, b, c, ...).*

Rule 1.1.2. ***fission*** *it show us how to break an obj. (this rule is scheme it means it works everywhere for every obj.)*

$o.$		$\{a_0, a_1 ... a_n\}$
$m.$	a_0	
	$\vdots$	
$m+n.$	a_n	

Example 1.1.1. $\{a, b, c\} / \therefore \{a, b\}$ *(prove if $\{a, b, c\}$ exists then $\{a, c\}$ exists)*

1.		$\{a, b, c\}$	
2.	a		*fi1*
3.	b		*fi1*
4.	c		*fi1*
5.		$\{a, c\}$	*fu2,3,4*

Example 1.1.2. $\{a, b\}/ \therefore \{b, a\}$

1.		$\{a, b\}$	
2.	a		*fi1*
3.	b		*fi1*
4.		$\{b, a\}$	*fu2,3*

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Chapter 2

propositional logic (zeroth-order logic)

2.1 What is proposition?

definition 2.1.1. ***predicate*** *"set()" and identity "=" and membership "∈" symbols are predicates. "set()" is unary and identity and membership are binary.*

capital letter "(P, Q, R)" showes **predicates**.

definition 2.1.2. ***Atomic proposition*** *we define atomic propositions like this.*

1. *if "a" is an obj then "set(a)" is an atomic proposition.*
2. *if "a, b" are obj then "$a = b$" is an atomic proposition.*
3. *if "a, b" are obj then "$a \in b$" is an atomic proposition.*

we denote atomic propositions with "p_i".

Undefined 2.1.1. ***logical operators*** *$(\neg, \rightarrow, \wedge, \vee)$ tool that used between atoms to make more complex propositions (all of the opratores are binary except "$\neg$" that is unary).*

you can easily see that formula are made up by atomic propositions.in general we say formula is an atomic formula or combination of them.

definition 2.1.3. ***Propositions*** *these are propositions.*

1. *atomic propositions "(p_i)".*
2. *if "p" is a formula then "$\neg p$" is a formula too.*
3. *if "p, q" are two propositions then "$p \bigcirc q$" is a formula too.($\bigcirc$ can be "$\rightarrow, \wedge, \vee$").*

for example "$(p_0 \rightarrow p_1, \neg p_3)$"
we denote propositions with "$p, q, r, ...$".

2.2 Rules of propositions.

definition 2.2.1. ***proposition schema(variable proposition)*** *a proposition that is unknown for us and it works as variable.(we show them with "$\varphi, \psi, \theta, \ldots$")*

Undefined 2.2.1. ***Assumption for conditional proof(acp)*** *is a tool for assuming a proposition. (any proposition can't get out but it can get in. right line must be shoter than left one.(it can't fall out). and every hypothesis that opens must be closed.)*

$$\begin{array}{l|l|l|l}
1 & & & \\
2 & & \multicolumn{1}{|l}{} & \\
3 & & \multicolumn{1}{|l|}{} & \\
4 & & \multicolumn{1}{|l|}{} & \\
5 & & \multicolumn{1}{|l}{} & \\
6 & & &
\end{array}$$

Rule 2.2.1. ***rules of logical operations*** $(\to I, \to E, \wedge I, \wedge E, \vee I, \vee E, \neg I, \neg E, \perp I, \perp E)$ *this rule is scheme it means it works everywhere for every propositions. (thats why we use φ, ψ instead of p, q)*

$\begin{array}{\|l} \varphi \\ \vdots \\ \psi \end{array}$ $\therefore \varphi \to \psi \qquad \to I$	$\varphi \to \psi$ $\underline{\varphi}$ $\therefore \psi \qquad \to E$

Example 2.2.1. *for "p, q, r" we can prove these.*

1. $p \to q, q \to r / \therefore p \to r$
2. $q / \therefore p \to q$

$$\begin{array}{lll}
1 & p \to q & ass \\
2 & q \to r & ass \\
3 & \quad\mid p & acp \\
4 & \quad\mid q & \to E1,3 \\
5 & \quad\mid r & \to E2,4 \\
6 & p \to r & \to I
\end{array}$$

$$\begin{array}{lll}
1 & q & ass \\
2 & \quad\mid p & acp \\
3 & \quad\mid q & ass \\
4 & p \to q & \to I
\end{array}$$

$\dfrac{\varphi \quad \psi}{\therefore \varphi \wedge \psi}$ $\wedge I$	$\dfrac{\varphi \wedge \psi}{\therefore \varphi}$ $\wedge E$ $\qquad$ $\dfrac{\varphi \wedge \psi}{\therefore \psi}$ $\wedge E$

Example 2.2.2. *for "p, q, r" we can prove these.*

1. $p \wedge q / \therefore q \wedge p$

2. $(p \wedge q) \wedge r / \therefore p \wedge (q \wedge r)$

3. $p \to q / \therefore (p \wedge r) \to q$

4. $(p \wedge q) \to r / \therefore p \to (q \to r)$

5. $p \to (q \to r) / \therefore (p \wedge r) \to q$

1	$p \wedge q$	ass
2	p	$\wedge E, 1$
3	q	$\wedge E, 1$
4	$q \wedge p$	$\wedge I, 2, 3$

1	$(p \wedge q) \wedge r$	ass
2	$p \wedge q$	$\wedge E, 1$
3	r	$\wedge E, 1$
4	p	$\wedge E, 2$
5	q	$\wedge E, 2$
6	$q \wedge r$	$\wedge I, 3, 5$
7	$p \wedge (q \wedge r)$	$\wedge I, 4, 6$

1	$p \to q$	ass
2	$\quad\vert\ p \wedge r$	acp
3	$\quad\vert\ p$	$\wedge E, 2$
4	$\quad\vert\ q$	$\to E, 1, 3$
5	$(p \wedge r) \to q$	$\to I$

1	$(p \wedge q) \to r$	ass
2	$\quad\vert\ p$	acp
3	$\quad\vert\quad\vert\ q$	acp
4	$\quad\vert\quad\vert\ p \wedge q$	$\wedge I 2, 3$
5	$\quad\vert\quad\vert\ r$	$\to E 1, 4$
6	$\quad\vert\ q \to r$	$\to I$
7	$p \to (q \to r)$	$\to I$

1	$p \to (q \to r)$	ass
2	$\quad\vert\ p \wedge q$	acp
3	$\quad\vert\ p$	$\wedge E, 2$
4	$\quad\vert\ q \to r$	$\to E, 1, 3$
5	$\quad\vert\ q$	$\wedge E, 2$
6	$\quad\vert\ r$	$\to E, 4, 5$
7	$(p \wedge q) \to r$	$\to I$

$\dfrac{\varphi}{\therefore \varphi \vee \psi} \quad \vee I$ $\quad\quad$ $\dfrac{\varphi}{\therefore \psi \vee \varphi} \quad \vee I$	$\varphi \vee \psi$ $\mid \varphi$ $\mid \vdots$ $\mid \theta$ $\mid \psi$ $\mid \vdots$ $\mid \theta$ $\therefore \theta \quad\quad \vee E$

Example 2.2.3. *for "p, q, r" we can prove these.*

1. $p \vee q / \therefore q \vee p$
2. $(p \vee q) \vee r / \therefore p \vee (q \vee r)$
3. $p \wedge (q \vee r) / \therefore (p \wedge q) \vee (p \wedge r)$
4. $(p \to q) \wedge (r \to s) / \therefore (p \vee r) \to (q \vee s)$

1	$p \vee q$	*ass*
2	$\quad\mid p$	*acp*
3	$\quad\mid p \vee q$	$\vee I, 2$
4	$\quad\mid q$	*acp*
5	$\quad\mid q \vee p$	$\vee I, 4$
6	$q \vee p$	$\vee E, 1$

1	$(p \vee q) \vee r$	*ass*
2	$\quad\mid p \vee q$	*acp*
3	$\quad\mid\quad\mid p$	*acp*
4	$\quad\mid\quad\mid p \vee (q \vee r)$	$\vee I, 3$
5	$\quad\mid\quad\mid q$	*acp*
6	$\quad\mid\quad\mid q \vee r$	$\vee I, 5$
7	$\quad\mid\quad\mid p \vee (q \vee r)$	$\vee I, 6$
8	$\quad\mid p \vee (q \vee r)$	$\vee E, 2$
9	$\quad\mid r$	*acp*
10	$\quad\mid q \vee r$	$\vee I, 9$
11	$\quad\mid p \vee (q \vee r)$	$\vee I, 10$
12	$p \vee (q \vee r)$	$\vee E, 1$

1	$p \wedge (q \vee r)$	*ass*
2	$\quad p$	$\wedge E1$
3	$\quad q \vee r$	$\wedge E1$
4	$\quad\quad q$	*acp*
5	$\quad\quad p \wedge q$	$\wedge E2,4$
6	$\quad\quad (p \wedge q) \vee (p \wedge r)$	$\vee I5$
7	$\quad\quad r$	*acp*
8	$\quad\quad p \wedge r$	$I\wedge$ *2,7*
9	$\quad\quad (p \wedge q) \vee (p \wedge r)$	$\vee I8$
10	$\quad (p \wedge q) \vee (p \wedge r)$	$\vee I8$
11	$(p \wedge q) \vee (p \wedge r)$	$\vee E3$

1	$(p \to q) \wedge (r \to s)$	*ass*
2	$(p \to q)$	$\wedge E1$
3	$(r \to s)$	$\wedge E1$
4	$\quad p \vee r$	*acp*
5	$\quad\quad p$	*acp*
6	$\quad\quad q$	$\to E2,5$
7	$\quad\quad q \vee s$	$\vee I,6$
8	$\quad\quad r$	*acp*
9	$\quad\quad s$	$\to E3,8$
10	$\quad\quad q \vee s$	$\vee I9$
11	$\quad q \vee s$	$\vee E,4$
12	$(p \vee r) \to (q \vee s)$	$\to I$

definition 2.2.2. $\perp$ *:* $\varphi \wedge \neg\varphi$

φ $\vdots$ $\perp$ $\therefore \neg\varphi \qquad \neg I$	$\neg\varphi$ $\vdots$ $\perp$ $\therefore \varphi \qquad \neg E$

definition 2.2.3. ***schema of prove*** *you've got knowed by now what is schema of a rule. if you want to prove a theorem that it's schema form.you can't prove it directly. here we use a technique calls* ***"schema of prove"****. it let us prove for all situations at the same time.*

Theorem 2.2.1. (*schema*) $(\perp I, \perp E)$

φ $\neg\varphi$ $\therefore \perp \qquad \perp I$	$\perp$ $\therefore \varphi \qquad \perp E$

1	φ	*ass*
2	$\neg\varphi$	*ass*
3	$\varphi \wedge \neg\varphi$	$\wedge I$
4	$\perp$	$def.3$

1	$\perp$	*ass*
2	$\quad \neg\varphi$	*acp*
3	$\quad \vdots$	
4	$\quad \perp$	$rp1$
5	φ	$\neg E$

Attention 2.2.1. *Intuitionists can skip the "$\neg E$" rule that is the "RAA" rule in old notation and use "$\bot E$" as a rule instead.*

Example 2.2.4. *for "p, q, r" we can prove these.*

1. $p \to q, \neg q / \therefore \neg p$

2. $p \to q / \therefore \neg p \vee q$

1	$p \to q$	ass
2	$\neg q$	ass
3	$\quad\mid p$	acp
4	$\quad\mid q$	$\to E, 1, 3$
5	$\quad\mid \bot$	$\bot I, 2, 4$
6	$\neg p$	$\neg I$

1	$p \to q$	ass
2	$\quad\mid \neg(\neg p \vee q)$	acp
3	$\quad\mid\quad\mid p$	acp
4	$\quad\mid\quad\mid q$	$\to E, 1, 3$
5	$\quad\mid\quad\mid \neg p \vee q$	$\vee I, 4$
6	$\quad\mid\quad\mid \bot$	$\bot I, 2, 5$
7	$\quad\mid \neg p$	$\neg I$
8	$\quad\mid \neg p \vee q$	$\vee I, 7$
9	$\quad\mid \bot$	$\bot I, 2, 8$
10	$\neg p \vee q$	$\neg I$

definition 2.2.4. $\varphi \leftrightarrow \psi$: $(\varphi \to \psi) \wedge (\psi \to \varphi)$

Theorem 2.2.2. $(schema)$ $(\leftrightarrow I, \leftrightarrow E)$

$\mid \varphi$ $\mid \vdots$ $\mid \psi$ $\mid \psi$ $\mid \vdots$ $\mid \varphi$ $\therefore \varphi \leftrightarrow \psi \qquad \leftrightarrow I$	$\dfrac{\varphi \leftrightarrow \psi \quad \varphi}{\therefore \psi} \; \leftrightarrow E \qquad \dfrac{\varphi \leftrightarrow \psi \quad \psi}{\therefore \varphi} \; \leftrightarrow E$

1	$\mid \varphi$	acp
	$\mid \vdots$	
n	$\mid \psi$	
$n+1$	$\varphi \to \psi$	$\to, I$
$n+2$	$\mid \psi$	acp
	$\mid \vdots$	
m	$\mid \varphi$	
$m+1$	$\psi \to \varphi$	$\to, I$
$m+2$	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$\wedge I$
$m+3$	$\varphi \leftrightarrow \psi$	$def., m+3$

1	$\varphi \leftrightarrow \psi$	ass
2	φ	ass
3	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$def.1$
4	$\varphi \to \psi$	$\wedge E3$
5	ψ	$\to E1,4$

Exercise 2.2.1. *for "p, q, r" prove.*

1. $p / \therefore p$
2. $/ \therefore p \vee \neg p$
3. $\neg p, p \vee q / \therefore q$
4. $\neg p / \therefore p \to q$
5. $p \to q \therefore / \therefore \neg q \to \neg p$
6. $(p \vee q) \wedge (p \vee r) \therefore / \therefore (p \vee (q \wedge r))$
7. $\neg p \wedge \neg q \therefore / \therefore \neg(p \vee q)$
8. $\neg p \vee \neg q \therefore / \therefore \neg(p \wedge q)$
9. $(\neg p \to q) / \therefore ((p \to q) \to q)$
10. $/ \therefore (p \wedge q \to r) \wedge (p \wedge s \to r) \to (p \wedge (q \vee s) \to r)$
11. $/ \therefore (p \wedge q \to r) \vee (p \wedge s \to r) \to (p \wedge (q \wedge s) \to r)$
12. $/ \therefore (p \to (q \to r)) \to ((p \to q) \to (p \to r))$
13. $/ \therefore (p \to q) \to ((q \to r) \to (p \to r))$

Attention 2.2.2. *we can rewrite* ***Example*** *and* ***Exercise*** *propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

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Chapter 3

first-order logic

3.1 what is formula?

obj are two types.we know them and we know what exactly are we talking about and unknowns are mystery.all the obj we have used until now were known ones we call them **"constants"**.

Undefined 3.1.1. ***variables*** *A variable is a symbol (usually a letter like* $x, y, or z$*) that represents an unknown obj.*

definition 3.1.1. ***Atomic formula*** *we define atomic formulas like this.*

1. *if "*x*" is an obj(constant or variable) then "*$set(x)$*" is an atomic formula.*
2. *if "*x, y*" are obj(constant or variable) then "*$x = y$*" is an atomic formula.*
3. *if "*x, y*" are obj(constant or variable) then "*$x \in y$*" is an atomic formula.*

we denote atomic formulas with "A_i*".*

definition 3.1.2. ***well formed formula(WWF)*** *these are formulas.*

1. *atomic formulas* (A_i)*.*
2. *if* A *is a formula then* $\neg A$ *is a formula too.*
3. *if* A, B *are two formulas then* $A \bigcirc B$ *is a formula too.(*$\bigcirc$ *can be* $\rightarrow, \wedge, \vee$*).*

for example $(A_0 \rightarrow A_1, \neg A_3)$
we easily call WWF "formula" and also we denote WWF with $A, B, C, \ldots$.

definition 3.1.3. ***open formula*** *formulas that variables used in one of its atomic formulas.*

the formulas that are not open call **"closed formulas"** or **"sentence"**. Now question is "are very closed formulas proposition?" (propositions in classic logic are sentences that are true or false?) the answer is unfortunately **No**.

definition 3.1.4. ***Substitution for obj*** *we define Substitution an obj with variable like this.*

1. *for variables if* $x = y$ *then* $y[a/x] := a$ *and if* $x \neq y$ *then* $y[a/x] := y$*.*

2. *for constants* $c[a/x] := c$.

definition 3.1.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single object we do it as we explain in "**Substitution for obj**". if it's like $x = y$ or $x \in y$ (x, y can be both constants and variables). we define it like this.*

$$set(y)[a/x] := set(y[a/x])$$

$$(x = y)[a/z] := (x[a/z] = y[a/z])$$

$$(x \in y)[a/z] := (x[a/z] \in y[a/z])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[a/x] := (B[a/x] \bigcirc C[a/x])$$

($\bigcirc$ can be $\to, \wedge, \vee$).

$$(\neg B)[a/x] := (\neg B[a/x])$$

simply means in a formula put an obj like a instead of a variable like x.we denote it by $A[a/x]$ or $A[\frac{a}{x}]$

definition 3.1.6. ***Open proposition*** *assume formula "A" with variables "$x_0, x_1, .., x_n$" in it and a set "U". we say formula "A"is a "open proposition in U" when for every obj "$a_0, a_1, .., a_n$" in "U"*

$$A([a_0/x_0][a_1/x_1]...[a_n/x_n])$$

is a proposition we call it open proposition and denote it with "$(P(x), Q(x, y), ..)$"

definition 3.1.7. ***Big "AND" and "OR"*** *assume a set like "$\{a_0, a_1, a_2, ..., a_n\}$" we define **big "AND"** like this.*

$$\bigwedge_{i=0}^{n} P(a_i) : P(a_0) \wedge P(a_1) \wedge P(a_2) \wedge ... \wedge P(a_n)$$

*and **big "OR"***

$$\bigvee_{i=0}^{n} P(a_i) : P(a_0) \vee P(a_1) \vee P(a_2) \vee ... \vee P(a_n)$$

definition 3.1.8. ***Universe of discourse*** *we specify a set like "$\{a_0, a_1, a_2, ..., a_n\}$" as universe of a Logic world.*

Rule 3.1.1. ***use of open propositions*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$"*

$c \quad P(x)$

means :

$$\begin{array}{ll} c_0 & P[a_0/x] \\ c_1 & P[a_1/x] \\ c_2 & P[a_2/x] \\ & \vdots \\ c_n & P[a_n/x] \end{array}$$

if Px was been assemed then like this

$$\begin{array}{l|l} c & P(x) \\ & \vdots \end{array}$$

means :

$$\begin{array}{l|l} c_0 & P[a_0/x] \\ & \vdots \\ c_1 & P[a_1/x] \\ & \vdots \\ c_2 & P[a_2/x] \\ & \vdots \\ c_n & P[a_n/x] \\ & \vdots \end{array}$$

definition 3.1.9. ***predicate schema(variable predicate)*** *a predicate that is unknown for us and it works as variable.(we show them with $\phi, \psi, \theta, ...$)*

definition 3.1.10. ***Universal quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define $\forall x P(x)$*

$$\forall x \phi(x) : \bigwedge_{i=0}^{n} \phi[a_i/x]$$

definition 3.1.11. ***Existential quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define $\exists x P x$*

$$\exists x \phi(x) : \bigvee_{i=0}^{n} \phi[a_i/x]$$

as you can see $\forall x \phi(x)$ and $\exists x \phi(x)$ are sentences.

definition 3.1.12. ***Free variables*** *variables that are not bound by a quantifier. (you can easily see that every formula with no free variables are sentences) (previously we sayed that every sentece is not necessary a proposition)*

3.2 Rules of first order logic.

Theorem 3.2.1. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$ *assume universe of discourse is "*$\{a_0, a_1, a_2, ..., a_n\}$*"*

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x \phi(x)}$ $\quad \forall I$	*1. x is a variable.* *2. $\phi(x)$ must not be assumed.*
$\dfrac{\forall x \phi(x)}{\therefore \phi[a/x]}$ $\quad \forall E$	*1. a is an obj in universe of discourse.(it can be constant or variable.)*
$\dfrac{\phi[a/x]}{\therefore \exists x \phi(x)}$ $\quad \exists I$	*1. a is an obj in universe of discourse.(it can be conatant or variable.)*
$\exists x \phi(x)$ $\quad \vert\ \phi(x)$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta$ $\quad \exists E$	*1. x is a variable.* *2. $\phi(x)$ must not be assumed in previous lines.(except ones that are closed)* *3. x is not free in θ*

1	$\phi(x)$	ass
2	$\phi[a_0/x]$	$def.1$
	$\vdots$	
n	$\phi[a_n/x]$	$def.1$
$n+1$	$\bigwedge_{i=0}^{n} \phi[a_i/x]$	$\wedge I, 2, 3, ..., n+2$
$n+2$	$\forall x\phi(x)$	$def.n+3$

1	$\forall x\phi(x)$	ass
2	$\bigwedge_{i=0}^{n} \phi[a_i/x]$	$def.1$
3	$\phi[a/x]$	$\wedge E2$

1	$\phi[a/x]$	
2	$\bigvee_{i=0}^{n} \phi[a_i/x]$	$\vee I1$
3	$\exists x\phi(x)$	$def2$

1	$\exists x\phi(x)$		
2	$\bigvee_{i=0}^{n} \phi[a_i/x]$		
3		$\phi[a_0/x]$	acp
		$\vdots$	
n		θ	
$n+1$		$\phi[a_1/x]$	acp
		$\vdots$	
m		θ	
	$\vdots$		
l		$\phi[a_2/x]$	acp
		$\vdots$	
r		θ	
$r+1$	θ		$\exists E, 3$

Example 3.2.1. *for conatants P,Q,R we can prove these.Proof Is Left As An Exercise To The Reader.*😀

1. $\forall x(P(x) \wedge Q(x)) \therefore / \therefore \forall xP(x) \wedge \forall xQ(x)$
2. $\exists x(P(x) \vee Q(x)) \therefore / \therefore \exists xP(x) \vee \exists xQ(x)$
3. $\forall xP(x) \vee \forall xQ(x) / \therefore \forall x(P(x) \vee Q(x))$
4. $\exists x(P(x) \wedge Q(x)) / \therefore \exists xP(x) \wedge \exists xQ(x)$
5. $\forall x(P(x) \to Q(x)) / \therefore \forall xP(x) \to \forall xQ(x)$
6. $\exists xP(x) \to \exists xQ(x) / \therefore \exists x(P(x) \to Q(x))$
7. $\forall xP(x) \wedge Q \therefore / \therefore \forall x(P(x) \wedge Q)$
8. $\forall xP(x) \vee Q \therefore / \therefore \forall x(P(x) \vee Q)$
9. $\exists xP(x) \wedge Q \therefore / \therefore \exists x(P(x) \wedge Q)$

10. $\exists x P(x) \lor Q \therefore / \therefore \exists x(P(x) \lor Q)$
11. $Q \to \forall x P(x) \therefore / \therefore \forall x(Q \to P(x))$
12. $Q \to \exists x P(x) \therefore / \therefore \exists x(Q \to P(x))$
13. $\forall x P(x) \to Q \therefore / \therefore \exists x(P(x) \to Q)$
14. $\exists x P(x) \to Q \therefore / \therefore \forall x(P(x) \to Q)$
15. $\exists x \neg P(x) \therefore / \therefore \neg \forall x P(x)$
16. $\forall x \neg P(x) \therefore / \therefore \neg \exists x P(x)$

Exercise 3.2.1. *for conatants P,Q,R prove.*

1. $\exists x\Big(P(x) \land \exists y Q(y) \to \forall z R(z)\Big) / \therefore \forall y \forall z\Big(\forall x P(x) \land Q(y) \to R(z)\Big)$
2. $\forall x \exists y\Big(P(x) \land Q(y) \to \forall z R(z)\Big) / \therefore \forall z\Big(\exists x P(x) \land \forall y Q(y) \to R(z)\Big)$
3. $\forall x P(x) \to \exists y Q(y) / \therefore \exists x \exists y\Big(P(x) \to Q(y)\Big)$
4. $\forall x \forall z \exists y \exists w\Big(P(y,z) \lor Q(z) \to R(x,w)\Big) / \therefore \Big(\exists z \forall y P(y,z) \lor \exists z Q(z)\Big) \to \forall x \exists w R(x,w)$
5. $\exists x P(x) \lor \exists x\Big(Q(x) \land R(x)\Big) / \therefore \exists x\Big(P(x) \lor Q(x)\Big) \land \exists x\Big(P(x) \lor R(x)\Big)$
6. $\exists x \exists z \forall y\Big(P(x,y) \to Q(z) \lor R(x,y)\Big) / \therefore \forall x \exists y P(x,y) \to \Big(\exists z Q(z) \lor \exists x \exists z R(x,y)\Big)$
7. $\forall x \exists y\Big(\exists z P(x,y,z) \land Q(x,y)\Big) \lor \forall x \exists y \exists z\Big(P(x,y,z) \land R(x,y)\Big) / \therefore \forall x \exists y \exists z\Big(P(x,y,z) \land (Q(x,y) \lor R(x,y))\Big)$

Attention 3.2.1. *we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

Chapter 4

The beginning of set theory

4.1 Axioms of set theory

we emphasis before that set has no definition.but we can describe what set is with rules and axioms.

Axiom 4.1.1. ***Existence of universe of set theory*** *There exists a set V (we denote it as V.it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that)*

definition 4.1.1. ***Subset*** *a subset is a set where every element is also an element of another, larger set called the superset*

$$A \subseteq B : \forall x(x \in A \to x \in B)$$

Theorem 4.1.1. (*schema*) $(\subseteq I, \subseteq E)$

$\quad\vert\ x \in A$ $\quad\vert\ \vdots$ $\quad\vert\ x \in B$ $\therefore A \subseteq B \qquad \subseteq I$	$A \subseteq B$ $a \in A$ $\therefore a \in B \qquad \subseteq E$

1	$\vert\ x \in A$	acp
	$\vert\ \vdots$	
n	$\vert\ x \in B$	
$n+1$	$x \in A \to x \in B$	$I \to, 1, n$
$n+2$	$\forall x(x \in A \to x \in B)$	$\forall I, n+1$
$n+3$	$A \subseteq B$	$def.n+2$

1	$A \subseteq B$	ass
2	$a \in A$	ass
3	$\forall x(x \in A \to x \in A)$	$def.1$
4	$a \in A \to a \in B$	$\forall E, 4$
5	$a \in B$	$E \to 3, 5$

Example 4.1.1. $A \subseteq B, B \subseteq C / \therefore A \subseteq C$

1	$A \subseteq B$	*ass*
2	$B \subseteq C$	*ass*
3	$\quad\mid\ x \in A$	*acp*
4	$\quad\mid\ x \in B$	$\subseteq E, 1, 3$
5	$\quad\mid\ x \in C$	$\subseteq E, 2, 4$
6	$A \subseteq C$	$\subseteq I, 3, 5$

Axiom 4.1.2. ***Extensionality*** *Two sets are equal (are the same set) if they have the same elements.*

$$\forall X, Y(\forall x(x \in X \leftrightarrow x \in Y) \rightarrow X = Y)$$

so we can rewrite axiom of Extensionality like this.

$$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \rightarrow X = Y)$$

Theorem 4.1.2. (*schema*) $(I =, E =)$

$\begin{array}{l\|l} & x \in A \\ & \vdots \\ & x \in B \\ \hline & x \in B \\ & \vdots \\ & x \in A \end{array}$ $\therefore A = B \qquad I =$	$\dfrac{\begin{array}{c} A = B \\ a \in A \end{array}}{\therefore a \in B} \quad E = \qquad \dfrac{\begin{array}{c} A = B \\ a \in B \end{array}}{\therefore a \in A} \quad E =$

1	$\quad x \in A$	*acp*
	$\quad \vdots$	
n	$\quad x \in B$	
$n+1$	$A \subseteq B$	$\subseteq I, 1, n$
$n+2$	$\quad x \in B$	*acp*
	$\quad \vdots$	
m	$\quad x \in A$	
$m+1$	$B \subseteq A$	$\subseteq I, n+2, m$
$m+2$	$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \rightarrow X = Y)$	*Extensionality*
$m+3$	$(A \subseteq B \wedge B \subseteq A) \rightarrow A = B$	$\forall E, m+2$
$m+4$	$A \subseteq B \wedge B \subseteq A$	$\wedge I, n+1, m+1$
$m+5$	$A = B$	$E \rightarrow, m+3, m+4$

Axiom 4.1.3. ***Weak comprehension*** *(schema) this axioms says for every open proposition with just one variable like $\phi(x, x_0, x_1, ..., x_n)$ there is a set such that every x is a member of that set if and only if it holds $\phi(x, x_0, x_1, ..., x_n)$.*

$$\forall x_0, x_1, ..., x_n \exists X \forall x(\phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X)$$

by axiom of extentionality we can show that X is unique. so we denote it as "$\{x|\phi(x, x_0, x_1, ..., x_n)\}$".

this axiom show that why Russell's paradox doesn't happened here because if ou assume "$R = \{x|x \notin x\}$" is a set.you can't prove or disprove "$R \in R$". (we don't need to prove that now.we will prove it in "**'MetaLogic'**".)

4.2 definitions of set theory

Let's talk about some definitions in set theoty.These definitions will come in handy later.

definition 4.2.1. ***Empty set*** *also known as the null set, is the unique mathematical set containing no elements at all. (uniqueness can be proven.)*

$$\varnothing := \{x \mid \bot\}$$

therefore.

$$\forall x(x \in \varnothing \leftrightarrow \bot)$$

therefore.

$$\forall x(x \notin \varnothing)$$

Example 4.2.1. $/ \therefore \forall X(\varnothing \subseteq X)$

$$
\begin{array}{lll}
1 & \quad\vert\ x \in \varnothing & acp \\
2 & \quad\vert\ \forall x(x \notin \varnothing) & def \\
3 & \quad\vert\ x \notin \varnothing & \forall E, 2 \\
4 & \quad\vert\ \bot & \bot I, 1, 3 \\
5 & \quad\vert\ x \in X & \bot E, 4 \\
6 & \varnothing \subseteq X & \subseteq I, 1, 5 \\
7 & \forall X(\varnothing \subseteq X) & \forall I, 6
\end{array}
$$

Lemma 4.2.1. $A = \varnothing \therefore / \therefore \forall y(y \notin A)$

$$
\begin{array}{lll}
1 & \forall y(y \notin A) & ass \\
2 & \quad\vert\ y \in A & acp \\
3 & \quad\vert\ y \notin A & \forall E, 1 \\
4 & \quad\vert\ \bot & \bot I, 2, 3 \\
5 & \quad\vert\ y \in \varnothing & \bot E, 4 \\
6 & \quad\vert\ y \in \varnothing & acp \\
7 & \quad\vert\ \forall y(y \notin \varnothing) & def. \\
8 & \quad\vert\ y \notin \varnothing & \forall E, 7 \\
9 & \quad\vert\ \bot & \bot I, 6, 8 \\
10 & \quad\vert\ y \in A & \bot E, 9 \\
11 & A = \varnothing & I =
\end{array}
$$

$$
\begin{array}{lll}
1 & A = \varnothing & ass \\
2 & \forall y(y \notin \varnothing) & def. \\
3 & \forall y(y \notin A) & = E, 1, 2
\end{array}
$$

Theorem 4.2.1. $(schema)$ $(I\varnothing \neq, E\varnothing \neq, \varnothing = I, \varnothing = E)$

$\dfrac{a \in A}{\therefore A \neq \varnothing}\quad I\varnothing \neq$	$\begin{array}{ll} A \neq \varnothing & \\ \quad\vert\ x \in A & \\ \quad\vert\ \vdots & \\ \quad\vert\ \theta & \\ \therefore \theta & E\varnothing \neq \end{array}$

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$$\begin{array}{l|l} & x \in A \\ & \vdots \\ & \bot \end{array}$$

$\therefore A = \varnothing \qquad \varnothing = I$

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$$\frac{A = \varnothing}{\therefore a \notin A} \qquad \varnothing = E$$

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1	$a \in A$	*ass*
2	$\exists x(x \in A)$	$\exists I, 1$
3	$\neg(\forall x(x \notin A))$	*De Morgan* 2
4	$A \neq \varnothing$	*lemma* 4.4.1, 3

1	$\quad\mid x \in A$	*acp*
	$\quad\mid \vdots$	
n	$\quad\mid \bot$	
$n+1$	$x \notin A$	$\neg I$
$n+2$	$\forall x(x \notin A)$	$\forall I, n+1$
$n+3$	$A = \varnothing$	*lemma* 4.4.1, 1

1	$A = \varnothing$	*ass*
2	$\forall x(x \notin A)$	*lemma* 4.4.1, 1
3	$a \notin A$	$\forall E2$

definition 4.2.2. ***pairing set***

$$\{a_0, a_1, a_2, ..., a_n\} := \{x \mid x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n\}$$

therefore.

$$\forall x(x \in \{a_0, a_1, a_2, ..., a_n\} \leftrightarrow (x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n))$$

Theorem 4.2.2. (*schema*) $(I\{\}, E\{\})$

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$$\frac{a = a_i}{\therefore a \in \{a_0, a_1, a_2, ..., a_n\}} \qquad I\{\}$$

</td>
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$a \in \{a_0, a_1, a_2, ..., a_n\}$

$$\begin{array}{l|l} & a = a_0 \\ & \vdots \\ & \theta \end{array}$$

$\vdots$

$$\begin{array}{l|l} & a = a_n \\ & \vdots \\ & \theta \end{array}$$

$\therefore \theta \qquad E\{\}$

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Proof Is Left As An Exercise To The Reader.😀

definition 4.2.3. ***Union and Intersection***
Intersection:

$$A \cap B := \{x \mid x \in A \land x \in B\}$$

therefore.

$$\forall x(x \in A \cap B \leftrightarrow (x \in A \land x \in B))$$

Union:

$$A \cup B := \{x \mid x \in A \lor x \in B\}$$

therefore.

$$\forall x(x \in A \cup B \leftrightarrow (x \in A \lor x \in B))$$

Theorem 4.2.3. (*schema*) $(\cap I, \cap E, \cup I, \cup E)$

$\begin{array}{ll} a \in A & \\ \underline{a \in B} & \\ \therefore a \in A \cap B & \cap I \end{array}$	$\begin{array}{ll} \underline{a \in A \cap B} & \\ \therefore a \in A & \cap E \end{array} \qquad \begin{array}{ll} \underline{a \in A \cap B} & \\ \therefore a \in B & \cap E \end{array}$
$\begin{array}{ll} \underline{a \in A} & \\ \therefore a \in A \cup B & \cup I \\ \underline{a \in A} & \\ \therefore a \in B \cup A & \cup I \end{array}$	$\begin{array}{ll} a \in A \cup B & \\ \quad \left\lvert \begin{array}{l} a \in A \\ \vdots \\ \theta \end{array} \right. & \\ \quad \left\lvert \begin{array}{l} a \in B \\ \vdots \\ \theta \end{array} \right. & \\ \therefore \theta & \cup E \end{array}$

Proof Is Left As An Exercise To The Reader.😀

definition 4.2.4. ***Big Union and Intersection***
Intersection:

$$\bigcap A := \{x \mid \forall X(X \in A \to x \in X)\}$$

i.e.

$$\bigcap A := \{x \mid \forall X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcap A \leftrightarrow \forall X(X \in A \to x \in X)\Big)$$

Union:

$$\bigcup A := \{x \mid \exists X(X \in A \land x \in X)\}$$

i.e.

$$\bigcup A := \{x \mid \exists X \in A \quad x \in X\}$$

therefore.

$$\forall x(x \in \bigcup A \leftrightarrow (\exists X(X \in A \land x \in X)))$$

Theorem 4.2.4. (*schema*) ($\bigcap I, \bigcap E, \bigcup I, \bigcup E$)

$\left\lvert \begin{array}{l} X \in A \\ \vdots \\ a \in X \end{array} \right.$ $\therefore a \in \bigcap A \qquad \bigcap I$	$a \in \bigcap A$ $\underline{B \in A}$ $\therefore a \in B \qquad \bigcap E$
$a \in B$ $\underline{B \in A}$ $\therefore a \in \bigcup A \qquad \bigcup I$	$a \in \bigcup A$ $\left\lvert \begin{array}{l} a \in X \\ X \in A \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \qquad \bigcup E$

Proof Is Left As An Exercise To The Reader.😀

definition 4.2.5. ***Complement***

$$A^c := \{x \mid x \notin A\}$$

therefore.

$$\forall x(x \in A^c \leftrightarrow x \notin A)$$

Theorem 4.2.5. (*schema*) ($()^c I, ()^c E$)

$\left\lvert \begin{array}{l} a \in A \\ \vdots \\ \bot \end{array} \right.$ $\therefore a \notin A \qquad ()^c I$	$\left\lvert \begin{array}{l} a \notin A \\ \vdots \\ \bot \end{array} \right.$ $\therefore a \in A \qquad ()^c E$

Proof Is Left As An Exercise To The Reader.😀

Observation 4.2.1. *we can show Universal set with...*

$$V = \{x \mid \top\}$$

therefore.

$$\forall x(x \in V \leftrightarrow \top)$$

therefore.

$$\forall x(x \in V)$$

definition 4.2.6. ***Subtraction***

$$A - B := \{x \mid x \in A \land x \notin B\}$$

therefore.

$$\forall x\Big(x \in A - B \leftrightarrow (x \in A \land x \notin B)\Big)$$

Theorem 4.2.6. (*schema*) $(-I, -E)$

$\dfrac{a \in A \qquad a \notin B}{\therefore a \in A - B} \quad -I$	$\dfrac{a \in A - B}{\therefore a \in A} \quad -E \qquad \dfrac{a \in A - B}{\therefore a \notin B} \quad -E$

Proof Is Left As An Exercise To The Reader.😀

Lemma 4.2.2. $/ \therefore A - B = A \cap B^c$

Proof Is Left As An Exercise To The Reader.😀

definition 4.2.7. ***Powerset***

$$P(A) := \{X \mid X \subseteq A\}$$

therefore.

$$\forall X\Big(X \in P(A) \leftrightarrow (X \subseteq A)\Big)$$

Theorem 4.2.7. (*schema*) (PI, PE)

$\dfrac{A \subseteq B}{\therefore A \in P(B)} \quad PI$	$\dfrac{A \in P(B)}{\therefore A \subseteq B} \quad PE$

Proof Is Left As An Exercise To The Reader.😀

Chapter 5

Relations and functions

5.1 cartesian product

definition 5.1.1. ***Ordered pair***

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

definition 5.1.2. ***Tuples****(n is a arbitrary number.)(this is a temporary definition, we will give a better one in the next chapter.)*

$$(x_1, x_2, ..., x_n) := ((x_1, x_2, ..., x_{n-1}), x_n)$$

Lemma 5.1.1. $/ \therefore (x_0, y_0) = (x_1, y_1) \leftrightarrow ((x_0 = x_1) \wedge (y_0 = y_1))$

Proof Is Left As An Exercise To The Reader.😀

definition 5.1.3. ***Cartesian product***

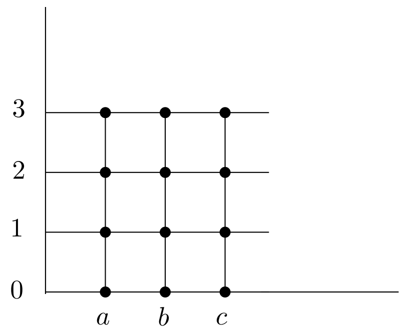
$$A \times B := \{u \mid \exists x, y\Big((x, y) = u \wedge x \in A \wedge y \in B\Big)\}.$$

or i.e.

$$A \times B := \{(x, y) \mid x \in A \wedge y \in B\}.$$

Example 5.1.1. *if* $A = \{a, b, c\}$ *and* $B = \{0, 1, 2, 3\}$ *then*

$$A \times B = \{(a, 0), (a, 1), (a, 2), (a, 3), (b, 0), (b, 1), (b, 2), (b, 3), (c, 0), (c, 1), (c, 2), (c, 3)\}$$



Exercise 5.1.1. *for sets A, B, C, prove.*

1. $/ \therefore A \times (B \cap C) = (A \times B) \cap (A \times C)$

2. $/ \therefore A \times (B \cup C) = (A \times B) \cup (A \times C)$

3. $/ \therefore A \times (B - C) = (A \times B) - (A \times C)$

5.2 Relations

definition 5.2.1. ***Relation*** *we call a set like* ***R*** *a* ***Relation*** *when there are two sets like X and Y s.t. $R \subseteq X \times Y$.*

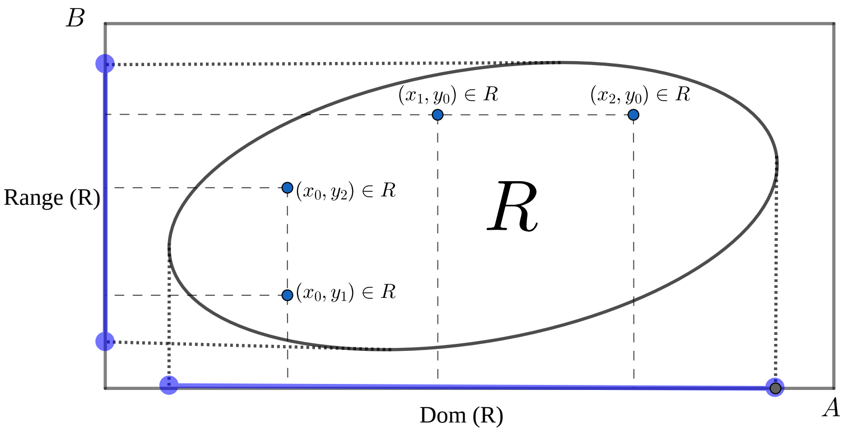
$$Rel(R) : \exists X, Y(R \subseteq X \times Y).$$

definition 5.2.2. ***Domain and Range(also called Image)***
Domain of a set R is all x that there exists a y s.t. "$(x, y) \in R$"

$$Dom(R) := \{x \mid \exists y \quad (x, y) \in R\}$$

Range(Image) of a set R is all y that there exists a x s.t. $(x, y) \in R$

$$Range(R) := \{y \mid \exists x \quad (x, y) \in R\}.$$



definition 5.2.3. ***Inverse of a Relation***

$$R^{-1} = \{u \mid \exists x, y\Big(u = (x, y) \wedge (y, x) \in R\Big)\}$$

or i.e.

$$R^{-1} = \{(x, y) \mid (y, x) \in R\}$$

therefore.

$$\forall x, y\Big((x, y) \in R^{-1} \leftrightarrow (y, x) \in R\Big)$$

definition 5.2.4. ***Combination of a Relations***

$$S \circ R = \{u \mid \exists x, y\Big(u = (x, y) \wedge \exists z\big((x, z) \in R \wedge (z, y) \in S\big)\Big)\}$$

or i.e.

$$S \circ R = \{(x, y) \mid \exists z\Big((x, z) \in R \wedge (z, y) \in S\Big)\}$$

therefore.

$$\forall x, y\Big((x, y) \in S \circ R \leftrightarrow \exists z\big((x, z) \in R \wedge (z, y) \in S\big)\Big)$$

definition 5.2.5. ***Image***

$$R[A] = \{y \mid \exists x \in A \quad (x, y) \in R\}$$

you can olso see by definition of "inverse of a Relation" and "image" this equation holds.

$$R^{-1}[B] = \{x \mid \exists y \in B \quad (x, y) \in R\}$$

*we cal this "**Pre-image**"*

5.3 Functions

definition 5.3.1. ***Function*** *we call a relation "**f**" a "**function**" when every input has exactly one output i.e.*

$$Func(f) : \forall x, y_1, y_2\Big((x, y_1) \in f \wedge (x, y_2) \in f \to y_1 = y_2\Big).$$

definition 5.3.2. ***Output of a function*** *assume a function **f** and $(x, y) \in f$, we define "$f(x) := y$" and we call "$f(x)$" a "**output of a function**". and the reverse is also true, if we use "$f(x)$", we have a function "**f**" and we can say "$x \in Dom(f)$". "output of a function" is object too, so we can have $f(f(x))$ as well.*

Attention 5.3.1. *input of a function can be a ordered pair itself, so we can have $f((x, y))$ and it is a "**output of a function**". we define "$f(x, y) := f((x, y))$" for simplicity. this is true for tuples as well, so we can have "$f((x_1, x_2, ..., x_n))$" and we define "$f(x_1, x_2, ..., x_n) := f((x_1, x_2, ..., x_n))$" for simplicity.*

definition 5.3.3. ***Term*** *we define term like this.*

1. *every obj is a term.*
2. *if $t_0, ...t_n$ are terms and "$Func(f)$" and "$(t_0, ...t_n) \in Dom(f)$". then "$f(t_0, ...t_n)$" is also a term.*

we denote terms with small letters like $t, s, r, ...$.

definition 5.3.4. ***Substitution for term*** *we define Substitution an term with variable like this.*

1. *for variables if $x = y$ then $y[t/x] := t$ and if $x \neq y$ then $y[t/x] := y$.*
2. *for constants $c[t/x] := c$.*
3. *for terms like $f(t_0, ...t_n)$*

$$f(t_0, ...t_n)[t/x] := f(t_0[t/x], ...t_n[t/x])$$

definition 5.3.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single term we do it as we explain in "**Substitution for term**". if it's like $t_0 = t_1$ or $t_0 \in t_1$ (t_0, t_1 are terms). we define it like this.*

$$set(y)[t/x] := set(y[t/x])$$

$$(t_0 = t_1)[t/x] := (t_0[t/x] = t_1[t/x])$$

$$(t_0 \in t_1)[t/x] := (t_0[t/x] \in t_1[t/x])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[t/x] := (B[t/x] \bigcirc C[t/x])$$

($\bigcirc$ can be $\to, \wedge, \vee$).

$$(\neg B)[t/x] := (\neg B[t/x])$$

Theorem 5.3.1. (*schema*) $(\forall I, \forall E, \exists I, \exists E)$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x\phi(x)}$ $\quad \forall I$	*1. x is a variable.* *2. "$\phi(x)$" must not be assumed.*
$\dfrac{\forall x\phi(x)}{\therefore \phi[t/x]}$ $\quad \forall E$	*1. t is a term.*
$\dfrac{\phi[t/x]}{\therefore \exists x\phi(x)}$ $\quad \exists I$	*1. t is a term.*
$\exists x\phi(x)$ $\quad\mid \phi(x)$ $\quad\mid \vdots$ $\quad\mid \theta$ $\therefore \theta$ $\quad \exists E$	*1. x is a variable.* *2. "$\phi(x)$" must not be assumed in previous lines.(except ones that are closed)* *3. x is not free in θ*

$(\forall I, \exists E)$ *is the same as* $(\forall I, \exists E)$ *in chapter three so we have to prove* $(\forall E, \exists I)$. *so assume* $t = f(x)$ *so to it's easy to see* $Func(f)$ *and* $x \in Dom(f)$.

1	$\forall x\phi(x)$	*ass*	1	$\phi(f(x))$	*ass*
2	$Func(f)$	*ass*	2	$Func(f)$	*ass*
3	$\exists y\big((x,y) \in f\big)$	*ass*	3	$\exists y\big((x,y) \in f\big)$	*ass*
4	$\quad\mid (x,y) \in f$	*acp*	4	$\quad\mid (x,y) \in f$	*acp*
5	$\quad\mid f(x) = y$	*def*	5	$\quad\mid f(x) = y$	*def*
6	$\quad\mid \phi(y)$	$\forall E$	6	$\quad\mid \phi(y)$	$= E$
7	$\quad\mid \phi(f(x))$	$= E$	7	$\quad\mid \exists x\phi(x)$	$\exists I$
8	$\phi(f(x))$	$\exists E$	8	$\exists x\phi(x)$	$\exists E$

Attention 5.3.2. *in all rules and theorems you can replace all constants like* $a, b, c, \ldots$ *with terms like* $t, s, r, \ldots.$ *and rewrite them.*

definition 5.3.6. ***Function from set to set*** *we call a function* ***f*** *from* ***A*** *to* ***B*** *when it's domain is* ***A*** *and it's range is subset of* ***B*** *i.e.*

$$(f : A \to B) : Func(f) \wedge Dom(f) = A \wedge Range(f) \subseteq B$$

definition 5.3.7. ***Injective function*** *we call a function* ***f*** *injective when every output has at* *most one input i.e.*

$$Injective(f) : \forall y, x_1, x_2\Big((x_1, y) \in f \wedge (x_2, y) \in f \to x_1 = x_2\Big).$$

*we call a function **f** injective from **A** to **B** when it's function from **A** to **B** and it's injective i.e.*

$$(f : A \xrightarrow{inj} B) : (f : A \to B) \wedge Injective(f)$$

definition 5.3.8. ***Surjective function*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is equal to **B** i.e.*

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge Range(f) = B$$

or i.e.

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge (\forall y \in B \quad \exists x \in A \quad (x, y) \in f)$$

definition 5.3.9. ***Bijection function*** *we call a function **f** from **A** to **B** when it's domain is **A** and it's range is equal to **B** i.e.*

$$(f : A \xrightarrow[surj]{inj} B) : (f : A \xrightarrow{inj} B) \wedge (f : A \xrightarrow[surj]{} B)$$

5.4 Family of sets

Attention 5.4.1. ***Indexes*** *are objects (like numbers) allows us to give a unique "tag" to every object. we use $i, j, k, ...$ for variable indexes and we use $m, n, l, ...$ for constant indexes.*

definition 5.4.1. ***Family of sets*** *assume $A : I \to \mathcal{F}$ is a function from a nonempty set of indices I to a family of sets $\mathcal{F}$, then we call $(Range[A])$ an indexed family of sets. we can write A_i instead of $A(i)$ and we can write $(A_i)_{i\in I}$ instead of "A" and $\{A_i\}_{i\in I}$ instead of "$Range[A]$" or "$A[I]$".*

$$(A_i)_{i\in I} := A$$

and

$$\{A_i\}_{i\in I} := A[I]$$

definition 5.4.2. ***Intersrction and union of an family of sets***

$$\bigcap_{i\in I} A_i := \bigcap \{A_i\}_{i\in I}$$

$$\bigcup_{i\in I} A_i := \bigcup \{A_i\}_{i\in I}$$

Attention 5.4.2. *we can write $\forall i \in I \quad P(i)$ instead of $\forall i(i \in I \to P(i))$ and we can write $\exists i \in I \quad P(i)$ instead of $\exists i(i \in I \wedge P(i))$.*

Lemma 5.4.1.

$$/ \therefore \bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$$

1	$x \in \bigcap_{i\in I} A_i$	acp
2	$x \in \bigcap\{A_i\}_{i\in I}$	$def.1$
3	$x \in \bigcap A[I]$	$def.1$
4	$\forall Y \in A[I] \quad x \in Y$	$def \bigcap$
5	$\forall Y((\exists i \in I \quad A_i = Y) \to x \in Y)$	$def image$
6	$(\exists i \in I \quad A_i = A_{i'}) \to x \in A_{i'}$	$\forall E$
7	$i' \in I$	acp
8	$A_{i'} = A_{i'}$	$= I$
9	$i' \in I \wedge A_{i'} = A_{i'}$	$\wedge I$
10	$\exists i \in I \quad A_i = A_{i'}$	$\exists I$
11	$x \in A_{i'}$	$\to E$
12	$i' \in I \to x \in A_{i'}$	$\to I$
13	$\forall i \in I \quad x \in A_i$	$\forall I$
14	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	def
15	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	acp
16	$\forall i \in I \quad x \in A_i$	def
17	$\exists i \in I \quad A_i = Y$	acp
18	$i \in I$	acp
19	$x \in A_i$	$\forall E$
20	$A_i = Y$	$\wedge E$
21	$x \in Y$	$= E$
22	$x \in Y$	$\exists E$
23	$(\exists i \in I \quad A_i = Y) \to x \in Y$	$\to I$
24	$\forall Y((\exists i \in I \quad A_i = Y) \to x \in Y)$	$\forall I$
25	$\forall Y \in A[I] \quad x \in Y$	def
26	$x \in \bigcap A[I]$	def
27	$x \in \bigcap\{A_i\}_{i\in I}$	def
28	$x \in \bigcap_{i\in I} A_i$	def
29	$\bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$	$= I$

$$/ \therefore \bigcup_{i\in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$$

1	$x \in \bigcup_{i\in I} A_i$	*acp*
2	$x \in \bigcup\{A_i\}_{i\in I}$	$def.1$
3	$x \in \bigcup A[I]$	$def.1$
4	$\exists Y \in A[I] \quad x \in Y$	$def \bigcup$
5	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$def.image$
6	$\quad x \in Y$	*acp*
7	$\quad \exists i \in I \quad A_i = Y$	*acp*
8	$\qquad i \in I$	*acp*
9	$\qquad A_i = Y$	*acp*
10	$\qquad x \in A_i$	$= E$
11	$\qquad \exists i \in I \quad x \in A_i$	$\exists I$
12	$\quad \exists i \in I \quad x \in A_i$	$\exists E$
13	$\exists i \in I \quad x \in A_i$	$\exists E$
14	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	def
15	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	*acp*
16	$\exists i \in I \quad x \in A_i$	def
17	$\quad i' \in I$	*acp*
18	$\quad x \in A_{i'}$	*acp*
19	$\quad A_{i'} = A_{i'}$	$= I$
20	$\quad \exists i \in I \quad A_i = A_{i'}$	$\exists I$
21	$\quad (\exists i \in I \quad A_i = A_{i'}) \wedge x \in A_{i'}$	$\wedge I$
22	$\quad \exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists I$
23	$\exists Y((\exists i \in I \quad A_i = Y) \wedge x \in Y)$	$\exists E$
24	$\exists Y \in A[I] \quad x \in Y$	def
25	$x \in \bigcup A[I]$	def
26	$x \in \bigcup\{A_i\}_{i\in I}$	def
27	$x \in \bigcup_{i\in I} A_i$	def
28	$\bigcup_{i\in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$	$= I$

Theorem 5.4.1. (*schema*) ($\bigcap_I I, \bigcap_I E, \bigcup_I I, \bigcup_I E$)

$\quad \vert\ i \in I$ $\quad \vert\ \vdots$ $\quad \vert\ t \in A_i$ $\therefore t \in \bigcap_{i \in I} A_i \qquad \bigcap_I I$	$t \in \bigcap_{i \in I} A_i$ $n \in I$ $\therefore t \in A_n \qquad \bigcap_I E$
$t \in A_n$ $n \in I$ $\therefore t \in \bigcup_{i \in I} A_i \qquad \bigcup_I I$	$t \in \bigcup_{i \in I} A_i$ $\quad \vert\ i \in I$ $\quad \vert\ t \in A_i$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta \qquad \bigcup_I E$

Proof Is Left As An Exercise To The Reader.😀